SPOTTER / ML ENGINEER ASSESSMENT

Freight rate prediction

A time-based regression solution for November-December 2025

The aim is to estimate the total posted price of a shipment from its load details and supplied market information. The solution uses January-October history, checks performance on later months, and produces the required final predictions.

Result at a glance

Historical backtest	MAE ($)	RMSE ($)	MAPE
Equipment-distance baseline	131.91	629.54	5.90%
Selected two-stage model	98.55	623.62	4.27%

Compared with the baseline, the selected model reduces mean absolute dollar error by **25.3%** and mean absolute percentage error by **27.6%**. These are historical development results; November-December accuracy is unknown because those target prices are not supplied.

Headline metrics average the three fold scores equally and include every test row, including flagged extreme prices. MAE is average absolute dollar error; MAPE averages each shipment's absolute percentage error; RMSE gives large errors more influence.

Why this approach

I separated changing market conditions from shipment-specific patterns. A market adjustment and a recent price level capture broad movement; XGBoost learns how distance, equipment, weight and location relate to price. I excluded quote_signal because its historical price relationship changes across months.

Data and final outputs

Item	Scope
Labeled development data	48,000 loads | January 1-October 31, 2025
Final prediction batch	12,000 loads | November 1-December 31, 2025
Final training population	47,323 loads after 677 training-only price exclusions
December chart	31 daily predictions for one fixed shipment

What the report establishes

The model improves on a simple benchmark, reproduces the saved outputs and passes the supplied file checks. The main unresolved risks are recent-month underprediction, extreme prices, and the assumption that the supplied future market features are available.

Reading guide: method and cleaning (p. 2), validation (p. 3), experiments (p. 4), December chart (p. 5), limitations and reproduction (p. 6), diagnostic appendices (pp. 7-8).

02 / DATA PREPARATION AND MODEL DESIGN

Data preparation and model design

Data issue	Treatment and reason
Negative weight: 292 historical; 145 final	Take the absolute value, treating these as sign errors. This is an assumption, not independently verified corruption.
Missing weight: 300 historical; 165 final	Keep NaN for XGBoost's native missing routing and add a weight_missing indicator.
Missing market_index: 374 historical; 249 final	Fill from the mean of observed values on the same date in that input batch. No supplied date is entirely missing.
Unusual posted prices	Exclude training RPM outside 0.6-1.6 times the equipment-distance-band-month median. Retain all rows in headline evaluation.
Final-only locations	Eight new cities touch 1,447 final loads. Use consistent supplied coordinates rather than city-ID encodings.

Target and model

The target is **log(posted_rate / distance)**. Working in price per mile accounts for route length; the log scale makes effects multiplicative. Predictions are converted back to total dollars, without a log-to-mean bias correction.

The market coefficient is estimated after five alternating demeaning sweeps over lane-equipment groups and calendar-week groups. These are additive group effects, not joint lane-by-week effects. The level adjustment then uses the last 13 weeks of lane-demeaned, market-adjusted training log prices. This estimates associations, not a causal market effect.

Total price = distance x exp(tree estimate + b x log(market_index) + recent-level shift)

Inputs and settings

12 tree inputs: log distance; three equipment indicators (Dry Van, Flatbed, Reefer); weight; weight_missing; pickup latitude/longitude; delivery latitude/longitude; and latitude/longitude differences. The separate market layer uses log(market_index).

XGBoost configuration	Value
Trees / learning rate / max depth	1,500 / 0.03 / 6
Minimum child weight / row and column subsampling	5 / 0.8 and 0.8
Tree method / random seed / threads	hist / 42 / all available
Loss / remaining parameters	Squared error on transformed target / library defaults

Raw date, load_id, quote_signal, the computed t_months feature, city/lane target premiums, and an extrapolated straight-line trend are not active tree inputs. No scaler is used. The simple baseline uses median training RPM per equipment and distance band, multiplied by shipment distance.

03 / VALIDATION AND HISTORICAL PERFORMANCE

Validation and historical performance

Each fold trains on earlier dates and predicts the next two calendar months. There is no gap and no retraining within a test window. This mirrors the final two-month prediction window; it is not a fixed two-month lead for every shipment.

Fold	Training dates (2025)	Raw / fitted rows	Test dates (2025)	Test rows
F1	01-01 to 06-30	28,806 / 28,406	07-01 to 08-31	9,671
F2	01-01 to 07-31	33,718 / 33,253	08-01 to 09-30	9,429
F3	01-01 to 08-31	38,477 / 37,944	09-01 to 10-31	9,523

All-row results by fold

Fold / model	MAE ($)	RMSE ($)	MAPE	Median AE ($)
F1 / baseline	134.36	630.13	5.91%	58.89
F1 / model	87.24	620.50	3.75%	23.31
F2 / baseline	127.81	620.27	5.75%	56.57
F2 / model	98.75	616.24	4.28%	30.42
F3 / baseline	133.55	638.21	6.03%	57.95
F3 / model	109.67	634.13	4.79%	37.38

F3 (September-October) is closest to the final period. The three folds average to $98.55 MAE and 4.27% MAPE for the selected model. Mean RMSE is $623.62; the mean of fold median absolute errors is $30.37.

What changed in later months

Month	Unique loads	MAE ($)	MAPE	Signed error ($)
2025-07	4,912	91.52	3.65%	-49.32
2025-08	4,759	82.93	3.85%	-19.29
2025-09	4,670	112.33	4.60%	-85.26
2025-10	4,853	107.11	4.97%	-73.43

Percentage error rises in the later months, and mean signed dollar error is negative: the model tends to underprice. On non-flagged F3 rows, mean signed dollar error is still -$45.61, so the bias is not explained only by extreme prices.

For monthly diagnostics, repeated shipments are reduced to the latest forecast origin: F1 for July, F2 for August, and F3 for September-October. This gives 19,194 unique loads from 28,623 fold prediction occurrences. The test windows overlap and were reused for development; there is no untouched labeled final test.

Filtered diagnostic fold-mean MAPE is 1.95%. It uses actual test prices to identify a retrospective subset, so it is not a headline estimate for all loads. Appendix tables retain this distinction.

04 / MODEL COMPARISONS AND ROBUSTNESS

Model comparisons and robustness

Model changes were judged on the same chronological folds. The goal was a useful improvement across periods, rather than the lowest score on one favorable month.

Comparison	All-row MAE ($)	All-row MAPE
Simple baseline	131.91	5.90%
Trees without market layer	102.14	4.39%
Trees with raw market/time inputs	123.64	5.33%
Selected market + recent level	98.55	4.27%
Selected model + quote_signal	107.61	4.82%
Selected model, flagged prices kept	129.80	5.65%

Bounded final experiments

Experiment	Finding and decision
8 / 13 / 17-week price level	All-row MAPE: 4.37% / 4.27% / 4.35%. Retain 13 weeks.
Fixed 50:50 recent/trend blend	MAPE improves to 4.20%, but filtered F1 worsens by 0.436 percentage points. Reject under the preset stability rule.
Six tree settings	Depth 4 at rate 0.03 gives 4.2646% versus 4.2713% for the retained model. Filtered improvement is only 0.0081 points; keep the original settings under the preset rule.

Acceptance required at least 0.05 percentage points lower filtered fold-mean MAPE, no filtered fold worsening above 0.3 points, and no worsening of all-row mean MAE or MAPE. This is a project decision rule, not a significance test. The retained model is not best on every metric; additional tree tuning shows diminishing returns.

New locations and extreme prices

City exclusion test	Selected test loads	Reference MAPE	Withheld MAPE
3 cities; seed 42	999	4.15%	4.17%
8 cities; seed 42	2,516	4.75%	4.99%

Both tests hide cities from January-August training and score the same September-October shipments touching them. The increases are modest, but each is one selected city sample and also reduces training size. There are only 21 naturally unseen routes in F3 under the original route definition; that sample is too small for a broad guarantee.

A fixed tail classifier trained on January-August and tested on September-October achieved AUC 0.5015 using model features plus quote_signal. This classifier found little discrimination; it does not prove extreme prices are inherently unpredictable.

05 / DECEMBER PREDICTIONS

December predictions

The fixed shipment is **Lexington to Fort Wayne, 360 miles, Dry Van, 32,000 lb**. Only its date changes. The figure below is the chart produced by the supplied score.py, rendered in monochrome for this report.

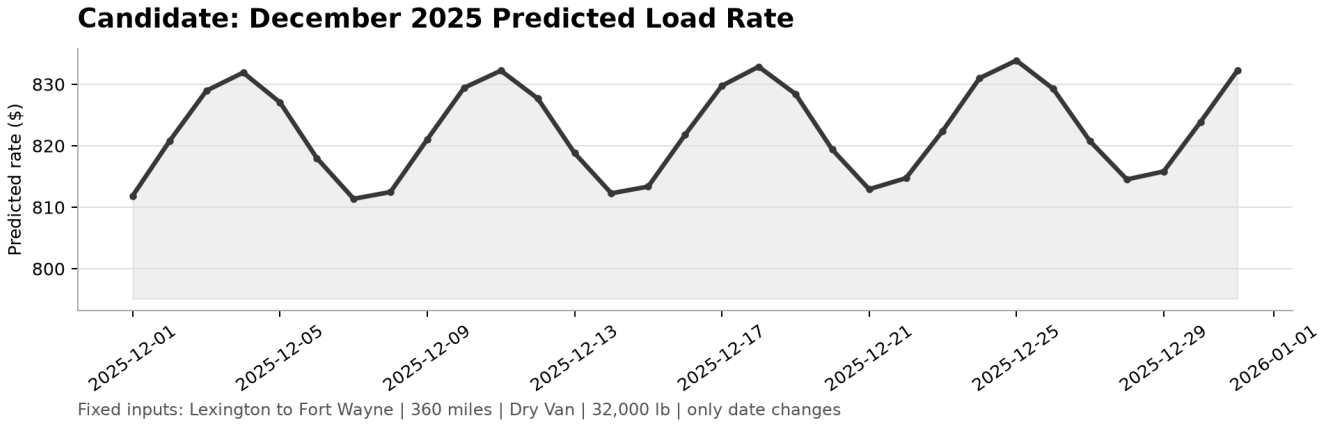


Figure 1. Submitted December 2025 predictions. Total prices span $811.37-$833.87, with a median of $821.77.

How the missing chart inputs are supplied

Coordinates come from the consistent city lookup built from filtered historical and unlabeled final feature rows. For each chart date, market_index is the mean of that date's final-batch observations. quote_signal is set to NaN and is not used. No final price labels are used.

This is a prediction conditional on the supplied November-December market features. It does not forecast an unknown future market index using historical data alone. The weekly wave comes entirely from those daily market means: the shipment's tree inputs are fixed, and no weekday or holiday feature is active.

Checks on the approximation and price level

September-October simulation	MAE ($)	MAPE
Normal row-level market inputs	109.6697	4.7870%
Daily mean market; quote_signal missing	110.0382	4.8003%
Change	+0.3685	+0.0132 points

This tests loss of row-level information on the same holdout dates, not the difficulty of forecasting future market values.

Lexington-Fort Wayne comparison	Rows	Price range ($)	Median RPM ($)
Historical route, all equipment	32	757.93-973.75	2.3241
Historical route, Dry Van only	21	757.93-934.37	2.2334
Fixed December predictions	31	811.37-833.87	2.2827

Historical distances and weights vary, so these are contextual comparisons rather than matching-load validation. The recent holdout has only six rows on this route. No separate December holiday effect can be verified from this history.

06 / LIMITATIONS, DELIVERY AND REPRODUCTION

Limitations, delivery and reproduction

What I would address next

Priority	Evidence and next step
1. Recent underpricing	Non-flagged signed percentage error is -2.25% in September and -1.71% in October. Test level calibration on new chronological origins before making further changes.
2. Extreme-price tail	277 flagged unique OOF rows contribute 55.54% of absolute dollar error. Investigate source-system price provenance and omitted commercial conditions; do not silently remove these rows from evaluation.
3. Future feature availability	The chart uses same-date final-batch market values. Confirm that availability contract before using this as an advance production forecast.
4. Validation uncertainty	The folds overlap and were reused for model selection. An independent future labeled period is needed for a less selection-dependent estimate.
5. Geographic coverage	City tests are encouraging but limited. Repeat fixed city exclusions with training-size controls before promising performance on every new location.

Reproducibility and output checks

Two fresh final fits produced identical unrounded predictions in the installed environment. After cent rounding, the reconstructed final and December outputs matched the saved files exactly. The supplied scorer accepted all 12,000 unique TE-000001 through TE-012000 IDs and all 31 fixed chart rows. Predictions are finite and positive.

The scorer validates structure and generates a figure; it does not calculate hidden accuracy. A clean-machine or cross-hardware reproduction has not been tested.

Run from the repository root

```
python -m pip install -r requirements-lock.txt

python -m src.eda

python -u -m src.validate

python -m src.predict

python score.py --predictions validation_predictions.csv
  --december-predictions data/december_chart_inputs.csv
```

The final two displayed lines form one scorer command. Prediction generation overwrites the two output CSVs. Verified runtime: Python 3.12.14; NumPy 2.5.4; pandas 2.3.3; scikit-learn 1.9.1; XGBoost 3.4.1.

Repository map and evidence

src/data.py: cleaning and price flags. **src/features.py**: tree inputs. **src/model.py**: market layer and XGBoost. **src/validate.py**: chronological backtests. **src/predict.py**: final fitting and December feature reconstruction.

Numerical evidence: reports/backtest_results.csv; reports/{city_holdout,blend_experiment,window_experiment,tuning_experiment}/; audit/{per_fold,monthly,breakdowns,december_simulation}.csv; audit/facts.json; audit/residual_tail/. Detailed audit tables remain in the repository. All report values come from these saved or reproduced outputs.

07 / APPENDIX A: ERROR CONCENTRATION

Appendix A: error concentration

Most recent fold: September-October, 9,523 loads. All rows included. Price buckets use actual target prices for retrospective diagnosis only; miles and cleaned pounds define the other buckets.

Group	Bucket	Loads	MAE ($)	MAPE
Equipment	Dry Van	5,360	106.23	4.45%
Equipment	Flatbed	1,770	110.28	5.38%
Equipment	Reefer	2,393	116.92	5.09%
Distance (mi)	<=250	550	21.73	4.50%
Distance (mi)	(250,500]	1,540	53.40	5.52%
Distance (mi)	(500,1000]	2,972	74.04	4.91%
Distance (mi)	(1000,2000]	2,927	148.51	4.57%
Distance (mi)	>2000	1,534	192.61	4.34%
Actual price ($)	<1000	1,545	67.79	13.28%
Actual price ($)	[1000,2000)	3,077	43.28	2.95%
Actual price ($)	[2000,4000)	3,387	72.77	2.55%
Actual price ($)	>=4000	1,514	369.90	4.86%
Weight quality	missing	65	82.25	3.19%
Weight quality	negative	59	103.72	9.57%
Weight quality	nonnegative	9,399	109.90	4.77%
Route	seen	9,502	109.80	4.79%
Route	unseen	21	50.21	2.11%

A seen route is an ordered pickup-delivery pair present before training outlier exclusion. If presence after exclusion is required, F3 has 22 unseen-route rows, MAE $51.07 and MAPE 2.14%. Small route and unusual-weight samples should be interpreted cautiously.

Non-flagged monthly residuals

Month	Loads	Signed error %	90th AE ($)	99th AE ($)
2025-07	4,847	-0.53%	88.99	182.56
2025-08	4,691	0.33%	76.35	168.56
2025-09	4,606	-2.25%	141.41	326.51
2025-10	4,773	-1.71%	104.97	213.33

Latest-origin predictions, one per shipment. Signed percentage error is the mean of (prediction - actual) / actual, not mean dollar error divided by mean price. Absolute-error percentiles use linear interpolation. These non-flagged rows are a target-conditioned diagnostic subset, not the headline population.

08 / APPENDIX B: SIGNAL AND CALENDAR CHECKS

Appendix B: signal and calendar checks

quote_signal changes relationship over time

Month	Mean	SD	Corr. RPM	Proxy MAE ($)	Proxy MAPE
2025-01	2.068	0.250	0.447	75.35	3.49%
2025-02	2.097	0.252	0.479	55.24	2.85%
2025-03	2.180	0.273	0.486	63.86	3.16%
2025-04	1.963	0.260	-0.472	403.31	20.10%
2025-05	1.913	0.266	-0.459	403.40	21.00%
2025-06	2.296	0.289	0.477	82.85	2.81%
2025-07	1.923	0.265	-0.468	415.77	20.68%
2025-08	2.052	0.223	-0.014	317.26	14.74%
2025-09	2.200	0.280	0.481	71.75	2.88%
2025-10	1.943	0.269	-0.434	413.72	20.99%
2025-11	2.055	0.220	N/A	N/A	N/A
2025-12	2.048	0.219	N/A	N/A	N/A

All rows; sample SD and Pearson correlation. Proxy = quote_signal x distance. N/A means NOT AVAILABLE because final target labels are absent. Raw correlations are affected by price extremes; the historical sign changes, not a claim of final independence, motivate exclusion.

Distribution shift and calendar checks

A classifier separating January-October from final feature rows achieved **AUC 0.7738** with deployed inputs, or **0.8194** after adding the excluded quote_signal. Complete dates were held out with seed 42; date, IDs and target fields were excluded. log(market_index) was the largest permutation-importance driver. This measures feature shift, not final regression accuracy.

After removing the fitted market component, non-flagged weekday mean RPM differs from its overall mean by -0.1584% to +0.2392%. These are descriptive comparisons without equipment or route controls. No weekday feature was added on this basis.

Holiday window (+/-3 days)	OOF loads	MAPE
Memorial Day: May 23-29	0	Not available
July 4: July 1-7	1,114	4.21%
Labor Day: Aug 29-Sep 4	1,052	3.03%

OOF forecasts begin July 1, so Memorial Day has no held-out coverage. These small windows do not identify a causal holiday premium and provide no direct December holiday validation. Final target accuracy, business error tolerance and a reliable estimate of gains from untested improvements remain unavailable.